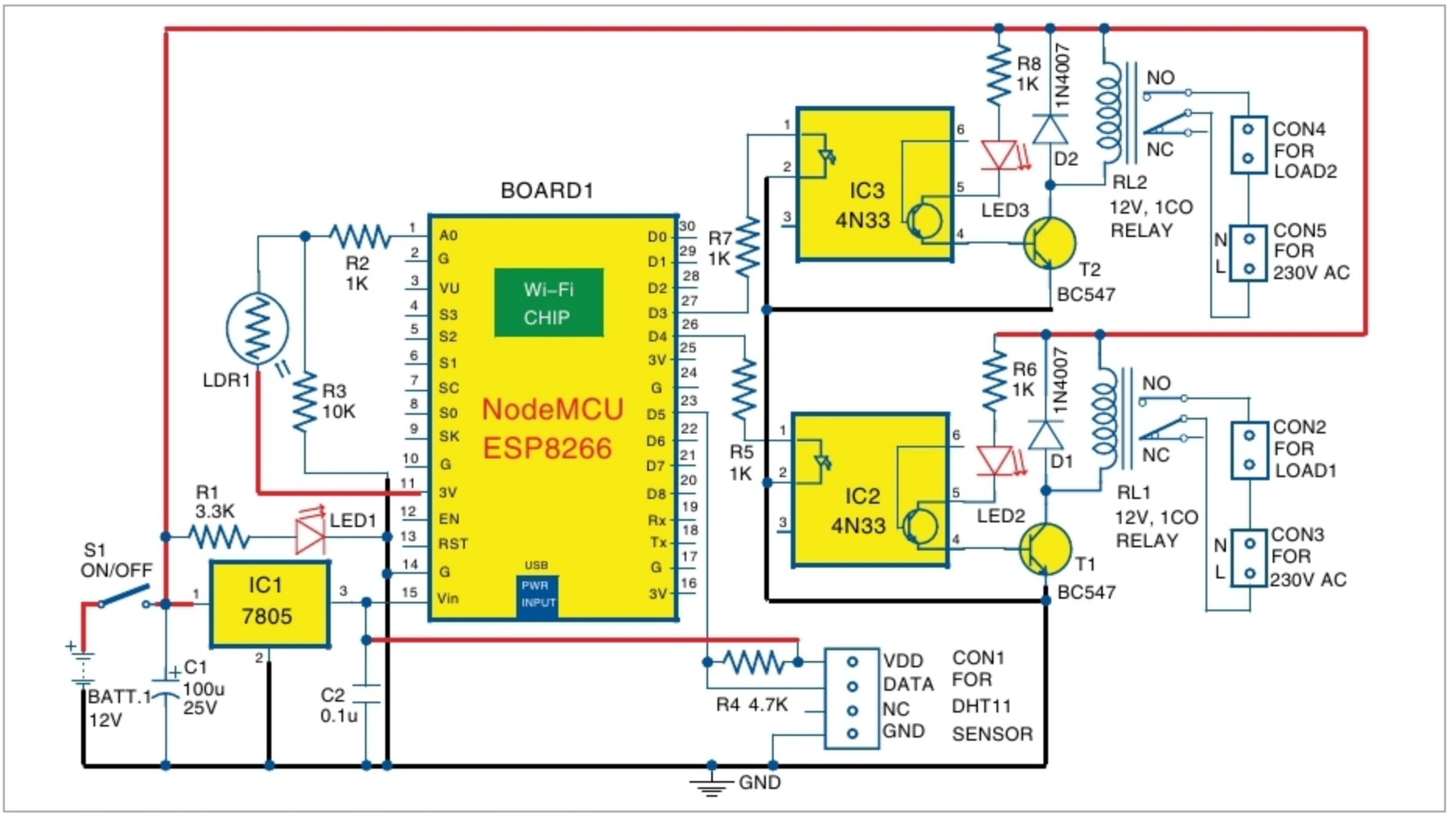


Fig. 3: Circuit diagram of ESP8266-based home automation system

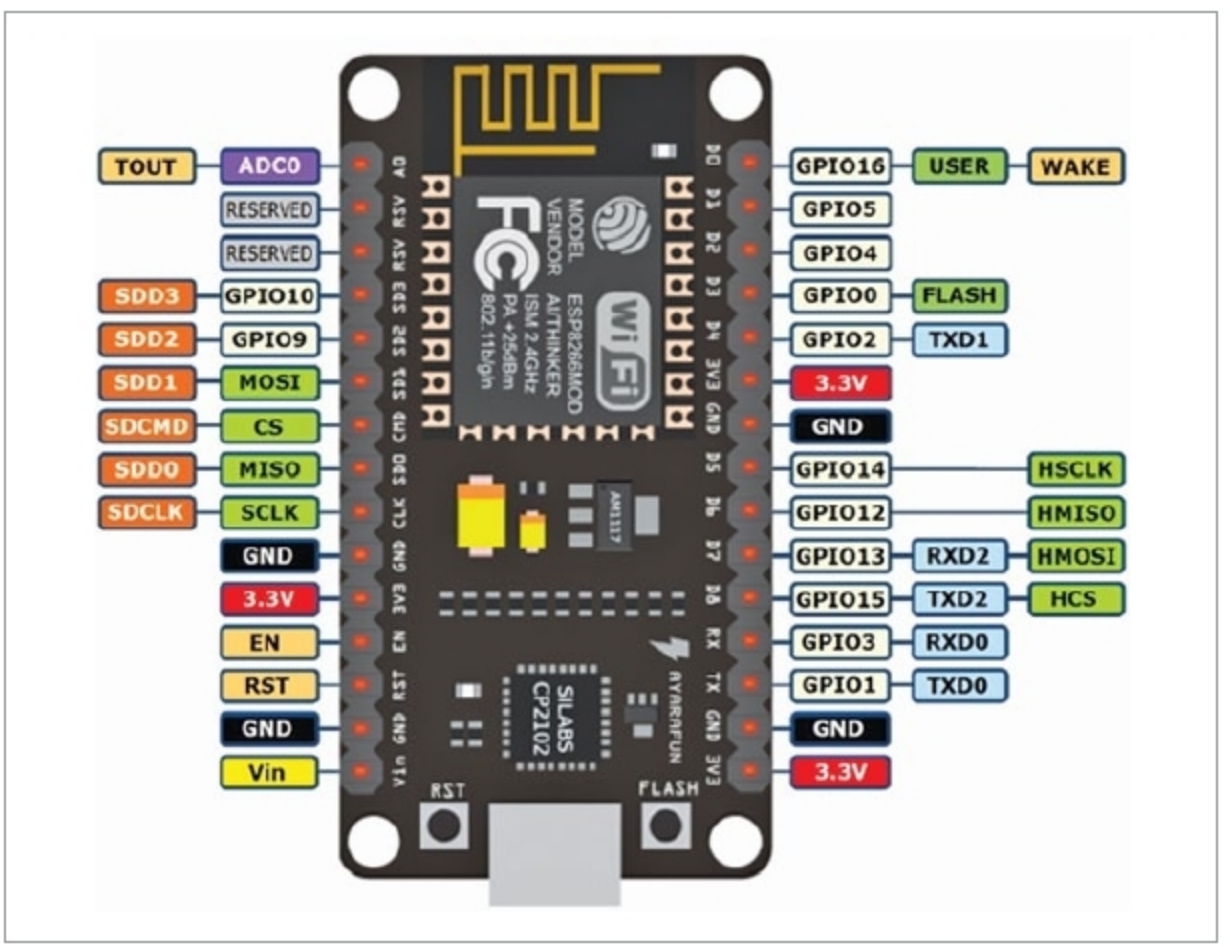


Fig. 4: NodeMCU module pin details

DHT11 VERSUS DHT22

Specification	DHT11	DHT22
RH range	20-80% (accuracy 5%)	0-100% (accuracy 2%)
Temperature range	0-50°C (accuracy 2°C)	-40 to 125°C (accuracy 0.5°C)
Sampling rate	1Hz	0.5Hz

through digital I/O pin D5 (GPIO14). This sensor can split the data for temperature and relative humidity, and send it one at a time through the same output pin.

Two 12V relays are driven by two BC547 transistors through MCT2E/4N33 opto-isolators. Each relay can drive AC/DC load through the terminal headers connected to it.

Software

The software (ESP8266_HA.ino) is written in Arduino programming language, which allows writing the code within a few lines. The program makes the device communicate with Cayenne platform when connected to the Internet through an access point or Wi-Fi router.

To add board for ESP8266 in Arduino IDE, go to→File→Preferences and paste the below link in Additional Board Manager URLs.

http://arduino.esp8266.com/stable/package_esp8266com_index.json

Open Boards Manager from Tools →Board menu and install esp8266 platform. (Do not forget to select your ESP8266 board from Tools→Board menu after installation.)